



MHF4U Unit 6: Rates of Change - Master Teaching Script

Slide 1: Title Slide Teacher Script: "Welcome back! Today we are officially starting Unit 6, which focuses on Rates of Change. This unit is the direct bridge between Advanced Functions and Calculus. Before we start the new material, we are going to do a rapid-fire review of Unit 5 to keep those skills sharp."

Slide 2: Unit 5 Recap - Identity vs. Equation Teacher Script: "In Unit 5, we dealt with two distinct types of problems. An 'Identity' is true for every single value of x in the domain. An 'Equation' is only true for specific values of x . Let's test your memory on solving both."

Slide 3: Unit 5 Practice 1 - Proving an Identity Teacher Script: "Here is your first review question. Prove the identity: $\tan(x) + \cot(x) = \sec(x)\csc(x)$. Grab your paper. What is the standard first rule when you are stuck on an identity?" Expected Answer: "Change everything to sine and cosine."

Slide 4: Unit 5 Practice 1 - Solution Teacher Script: "Exactly. By changing $\tan$ to $\sin/\cos$ and $\cot$ to $\cos/\sin$, we can find a common denominator. The numerator becomes $\sin^2(x) + \cos^2(x)$, which we know from the Pythagorean Identity equals 1. This simplifies perfectly to $\sec(x)\csc(x)$. LHS = RHS."

Slide 5: Unit 5 Practice 2 - Solving an Equation Teacher Script: "Excellent. Now let's solve an equation. Find all solutions for $2\cos^2(x) - 1 = 0$ on the interval $[0, 2\pi]$. What is your first algebraic step?" Expected Answer: "Isolate $\cos^2(x)$ to get $1/2$, then take the square root."

Slide 6: Unit 5 Practice 2 - Solution Teacher Script: "Correct. When you take the square root, do not forget the plus or minus! $\cos(x) = \pm 1/\sqrt{2}$. This means we have solutions in all four quadrants. Using the special $\pi/4$ triangle, we get $\pi/4$, $3\pi/4$, $5\pi/4$, and $7\pi/4$."

Slide 7: Transition to Unit 6 Teacher Script: "Your trigonometry foundation is solid. Let's leave the angles behind and enter Unit 6. Today, we are going to analyze how fast things move and how quickly functions grow."

Slide 8: What is a Rate of Change? Teacher Script: "A rate of change measures how one quantity changes in relation to another. If you drive 100 kilometers in 2 hours, your rate of change is 50 kilometers per hour. In algebra, the rate of change is simply the slope of a line."

Slide 9: Concept 1 - Average Rate of Change (AROC) Teacher Script: "There are two types of rates we care about. The first is the Average Rate of Change, or AROC. This measures the average speed over a specific interval, meaning we need exactly two distinct points on a graph."

Slide 10: Visualizing AROC - The Secant Line Teacher Script: "When we connect two points on a curve to find the AROC, we create a 'Secant Line'. The slope of this secant line represents the average rate of change. Notice how it cuts through the curve at exactly two places."

Slide 11: The AROC Formula Teacher Script: "The formula for AROC is the exact same slope formula you learned in Grade 9: y_2 minus y_1 , divided by x_2 minus x_1 . In function notation, it is written as $[f(x_2) - f(x_1)] / [x_2 - x_1]$."

Slide 12: Example 1 - AROC from a Table Teacher Script: "Let's look at this table showing the height of a plant over 4 weeks. Calculate the average rate of change between week 1 and week 4."

Slide 13: Example 1 - Solution Teacher Script: "Using the formula, we take the height at week 4 (12 cm) minus the height at week 1 (3 cm), divided by the time difference (4 - 1). This gives us $9 / 3$, which is 3 cm per week."

Slide 14: Example 2 - AROC from an Equation Teacher Script: "Now let's use an equation. Find the AROC of $f(x) = x^2 + 2x$ on the interval $[1, 3]$. First, calculate the y -values for $x=1$ and $x=3$."

Slide 15: Example 2 - Solution Teacher Script: "Evaluating the function, $f(1) = 3$ and $f(3) = 15$. The slope is $(15 - 3) / (3 - 1) = 12 / 2 = 6$. The average rate of change over this interval is 6."

Slide 16: The Limitation of AROC Teacher Script: "AROC is useful, but it has a massive limitation. If you drive from Toronto to Ottawa, your average speed might be 100 km/h. But were you driving exactly 100 km/h the entire time? No. You sped up, slowed down, and stopped at red lights."

Slide 17: Concept 2 - Instantaneous Rate of Change (IROC) Teacher Script: "This brings us to the Instantaneous Rate of Change, or IROC. This is what your speedometer reads at one exact, frozen moment in time. Not an interval, just one single point."

Slide 18: Visualizing IROC - The Tangent Line Teacher Script: "Visually, IROC is represented by a 'Tangent Line'. Unlike a secant line that cuts through two points, a tangent line perfectly skims the edge of the curve, touching it at only one specific coordinate."

Slide 19: The IROC Problem (Division by Zero) Teacher Script: "Here is the fundamental issue: to calculate a slope, we need two points. But IROC only happens at ONE point. If we try to use the slope formula on a single point (x_1, x_1) , the denominator becomes zero. We get $0/0$. So, how do we calculate it?"

Slide 20: The Solution - Estimating with Intervals Teacher Script: "We cheat. We pick our main point, and then we pick a second point that is incredibly close to it. So close that it is practically the same point. We usually use a microscopic horizontal distance of 0.001."

Slide 21: Preceding, Following, and Centered Intervals Teacher Script: "There are three ways to pick that second point. The preceding interval steps slightly backward $(x - 0.001)$. The following interval steps slightly forward $(x + 0.001)$. The centered interval averages both. In this course, we heavily focus on the following interval."

Slide 22: The Difference Quotient Formula Teacher Script: "This brings us to the Difference Quotient. It looks complicated, but it is literally just the slope formula adapted for a tiny distance, which we call 'h'. The formula is $[f(a + h) - f(a)] / h$."

Slide 23: IROC Calculation Setup Teacher Script: "Let's set this up. We want the IROC at $x = 3$. Using our formula, 'a' is 3. The step 'h' is 0.001. This means our two x-values are 3 and 3.001. The denominator is just 0.001."

Slide 24: Example 3 - IROC of a Quadratic Teacher Script: "Let's find the exact IROC of $f(x) = x^2$ at $x = 3$. Grab your calculator and calculate $f(3.001)$ very carefully."

Slide 25: Example 3 - Solution Teacher Script: " $f(3.001)$ is 9.006001. We know $f(3)$ is 9. Subtracting them gives 0.006001. Divide by 0.001, and you get exactly 6.001. We round this to the nearest integer, so the instantaneous rate of change is 6."

Slide 26: Example 4 - IROC of a Rational Function Teacher Script: "This method works for any function. Find the IROC of $f(x) = 1/x$ at $x = 2$. Set it up the exact same way using $h = 0.001$."

Slide 27: Example 4 - Solution Teacher Script: " $f(2.001)$ is $1 / 2.001$, which is roughly 0.49975. $f(2)$ is 0.5. When we subtract and divide by 0.001, we get -0.2498. We can confidently estimate the exact rate of change is -0.25, or $-1/4$."

Slide 28: Interpreting IROC: Positive, Negative, Zero Teacher Script: "Look at the sign of the answer. If the IROC is positive, the tangent line slopes up, meaning the function is increasing. If the IROC is negative, the function is decreasing. What happens if the IROC is exactly zero?"

Slide 29: Zero Rate of Change (Turning Points) Teacher Script: "If the tangent slope is exactly zero, the line is perfectly horizontal. This occurs at the exact peaks and valleys of a graph. These are called local maximums and minimums, or turning points."

Slide 30: Real-World Applications: Position and Velocity Teacher Script: "In physics and word problems, Unit 6 is primarily used to find velocity. The 'Position Function' gives height or distance.

The AROC of a position function gives you 'Average Velocity'. The IROC gives you 'Instantaneous Velocity'."

Slide 31: Word Problem: Falling Object Teacher Script: "A stone is dropped from a cliff. Its height in meters is given by $h(t) = 100 - 4.9t^2$. Find its instantaneous velocity at exactly $t = 3$ seconds."

Slide 32: Word Problem: Solution Teacher Script: "Set up the difference quotient with $a = 3$ and $h = 0.001$. $[h(3.001) - h(3)] / 0.001$. You get $(55.8705951 - 55.9) / 0.001$, which equals -29.4 m/s. The velocity is negative because the object is moving downward."

Slide 33: Summary Map: AROC vs IROC Teacher Script: "Let's lock this in. AROC requires two points, uses a secant line, and is an average over time. IROC requires one point, uses a tangent line, and is estimated using a tiny value for h like 0.001 ."

Slide 34: Homework & Next Steps Teacher Script: "Excellent work today. For homework, complete the Unit 6 Workbook problems on estimating IROC and Velocity. Next session, we will explore advanced word problems and graphical interpretations of rates. Have a great day!"